Module 1 Simulation: Destination Comparator

What this simulation does

The Destination Comparator lets you plant a crop in four environments and compare what happens to it in each one. The four destinations are Earth (used as a baseline), the Moon, Mars, and the ISS (International Space Station). The simulation walks you through four steps: Pick, Predict, Compare, and Reflect.

The goal is not to find the right answer. The goal is to observe what changes between destinations and use that to explain why Earth farming assumptions break down in space.

Step 1: Pick a Crop

You will see three crop options displayed as buttons. Currently, Lettuce is the fully active option. Tomato and Potato are placeholder options marked as coming soon — you can click them but they will not generate full outcome data yet.

Select Lettuce. A gold border will appear around it to confirm your selection. Click CONTINUE to move to the next step.

Step 2: Predict

You will see three off-Earth destinations: the Moon, Mars, and the ISS. Earth is already set as your baseline — you will not need to select it.

Read the brief description under each destination before choosing:

Moon: No atmosphere. 14-day nights.

Mars: Thin air. Dust storms. Cold.

ISS: Microgravity. No natural sun.

Select the destination where you think your crop will struggle the most. There is no penalty for being wrong — this prediction is here to make you commit to a hypothesis before you see the results. Click PLANT AND COMPARE to run the simulation.

Step 3: Compare (Results)

The results screen shows four cards, one for each destination. Each card displays:

The destination name and icon.

The outcome label: Baseline (Earth), Thriving, Struggling, or Failed.

A caption explaining what happened and why, written in plain language.

A variable focus label at the top of the screen identifies which environmental factor the simulation is highlighting for your crop. For Lettuce, the variable is Light.

Your prediction is shown above the result cards so you can compare what you expected against what actually happened. Read every caption, not just the destination you predicted. The differences between outcomes are where the learning is.

What the outcomes mean for Lettuce:

Earth (Baseline): Lettuce grows to full size with dark green leaves under natural sunlight and a predictable day-night cycle. This is the reference point for all other outcomes.

Moon (Failed): Each lunar night lasts about 14 Earth days. Without artificial grow lights, photosynthesis halts for half the cycle and the lettuce wilts and dies in the dark.

Mars (Struggling): Sunlight at Mars is roughly 43 percent of Earth intensity, and dust storms can block it for weeks. Leaves grow pale and stunted from chronic low light.

ISS (Thriving): No natural sunlight, but tuned red and blue LED grow lights replace it. NASA's Veggie experiments have grown real lettuce here. The heads are slightly smaller than Earth lettuce but healthy in color.

Click REFLECT to move to the final step.

Step 4: Reflect

You will see three text boxes. Write one or two sentences in each:

One condition that helped your crop: what part of the destination's environment let it grow?

One condition that limited or killed it: what stopped the plant from succeeding?

One question for Module 2: what do you want to understand about keeping a crop alive off-world?

These answers go directly into your Module 1 Discussion 1 post. You can click COPY MY REFLECTION to copy all three to your clipboard, then paste them into the discussion form. Click TRY A DIFFERENT CROP to start over with a new selection.

If you get stuck

Read each result caption out loud. The caption tells you the mechanism, not just the outcome. If the ISS outcome surprises you (the crop thrives with no natural sunlight), that surprise is the lesson: the variable is light availability, not sunlight specifically. Re-run with the same crop and a different prediction to see how your thinking changes.